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Housing Affordability Case Study
April 2026
Data Analysis Portfolio Sample

Introduction

Housing affordability varies widely across regions, and understanding the relationship between wages and rent is essential for evaluating how attainable housing truly is. This case study compares Sussex County, Delaware, a mixed rural and seasonal economy, with Philadelphia County, Pennsylvania, a large urban labor market. By examining wages, rent levels, and the number of work hours required to afford housing, this analysis highlights how local economic structures shape affordability in ways that are not always intuitive.

Data Sources

This analysis uses 2025 wage and rent data from the U.S. Census Bureau, the Bureau of Labor Statistics (BLS), and publicly available rental market datasets. Weekly wage data was converted to hourly wages using a standard 40-hour workweek. Hours needed to afford rent were calculated by dividing median monthly rent by median hourly wage for each region.

Comparison Table

Variable	Sussex County	Philadelphia County
Hourly wage	\$28.3	\$41.63
Median rent	\$1,238	\$1,397
Hours needed	43.7	33.5

Wage-Rent Gap Analysis

Philadelphia's median hourly wages, \$41.63, compared to \$28.30 in Sussex County, a difference of \$13.33. Rent, however, is only \$159 higher in Philadelphia, with Sussex at \$1,238 and Philadelphia at \$1,397. This means Philadelphia wages are roughly 41% higher, while rent is only about 13% higher. In other words, wage growth in Philadelphia far outpaces rent growth, creating a smaller wage-rent gap than in Sussex County. Even though rent is slightly higher in the city, the significantly higher wages improve overall affordability.

Hours Needed to Afford Rent

A worker in Sussex County must work about 43.7 hours per month just to cover rent. In Philadelphia, a worker needs only about 33.5 hours to afford a typical apartment. This means a full-time worker in Philadelphia has more income left over after paying rent, despite the city's higher housing costs. The combination of higher wages and a year-round labor market makes Philadelphia more affordable than a lower-wage, seasonal economy like Sussex County.

Economic Context: Seasonal vs. Year-Round Labor Markets

Sussex County's economy is shaped by seasonal industries such as tourism, hospitality, retail, and food service. These sectors tend to offer lower wages, fluctuating hours, and limited year-round stability. As a result, even modest rent levels can feel less affordable because workers face inconsistent income and fewer opportunities for higher-paying jobs.

Philadelphia, by contrast, operates on a year-round labor market anchored by hospitals, universities, government agencies, and professional services. These industries typically provide higher wages, steadier hours, and more predictable employment. This structure helps explain why Philadelphia workers can afford higher rent with fewer hours of labor, while Sussex County workers must devote more of their income to housing despite lower overall costs.

Conclusion

This analysis shows that housing affordability depends less on rent alone and more on the relationship between wages and housing costs. Although Philadelphia's rent is slightly higher, its significantly higher wages reduce the number of hours a worker must devote to housing each month. In Sussex County, lower wages and a seasonal labor market create a wider wage-rent gap, making even lower rent levels harder to sustain. Comparing these two regions highlights how local economic structures shape affordability and why wage strength is a critical factor in determining whether housing is truly attainable.

Sources

U.S. Census Bureau. "QuickFacts: Sussex County, Delaware; Philadelphia County, Pennsylvania." 2025.
Bureau of Labor Statistics (BLS). "Quarterly Census of Employment and Wages (QCEW)." 2025.
U.S. Census Bureau. "American Community Survey (ACS) 1-Year Estimates." 2025.
Zillow Research. "Zillow Observed Rent Index (ZORI)." 2025.